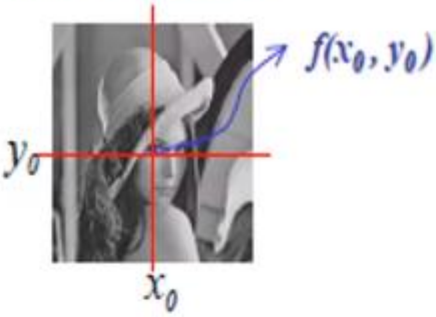
What is an Image?

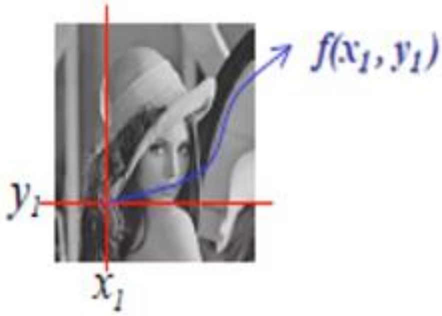
- Image is a representation of something or someone. Examples are any drawing, painting, photograph etc.
- Images are very powerful tool in communication.
- It may be black & white or color.

Often an image is defined as a two-dimensional function, f(x, y).

➢ where x and y are spatial (plane) coordinates

➢ the amplitude of f at any pair of coordinates (x, y) is called the intensity or gray level of the image at that point.





Pixel intensity value

$\underbrace{f(1,1)}_{\text{Pixel location}} = 103$

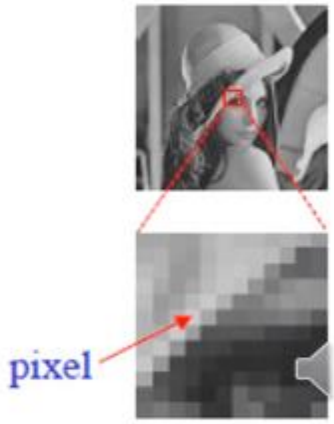
What is digital Image?

•When x, y, and f (x, y) are all finite, discrete quantities, we call the image a **digital image**.

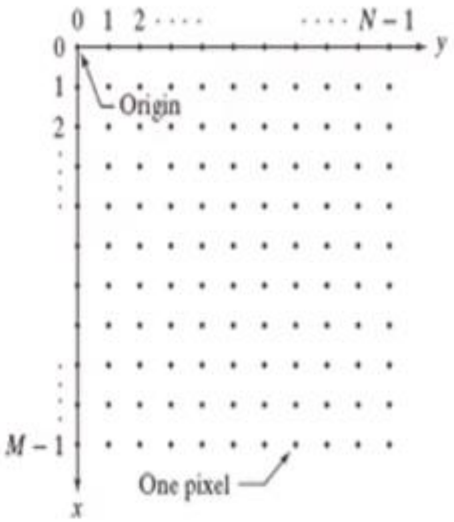
•Digital image is composed of a finite number of elements, each of which has a particular location and value. These elements are referred to as:

- **picture elements**
- **image elements**
- **Pels**
- **pixels**

•Pixel is the most widely used term.



- Digital image consists of *M* rows and N columns.
- Each pixels, each represented by *k* bits.
- A pixel can thus have 2^k different values.



A simple image model

- To be suitable for computer processing, an image f(x,y) must be digitalized both spatially and in amplitude.
- Digitization of the spatial coordinates (x,y) is called image sampling.
- Amplitude digitization is called gray-level quantization.

Remember digitization implies that a digital image is an approximation of a real scene

What is Image Processing?

- **Image processing is a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or to extract some useful information from it.**
- It is a type of **signal dispensation** in which input is image, and output may be image or characteristics associated with that image.
- It is among rapidly growing technologies today, with its applications in various aspects of a business. Image Processing forms core research area within engineering and computer science disciplines too.

Image processing basically includes the following three steps:

- Importing the image via image acquisition tool.
- Analyzing and manipulating the image
- Output is the last stage in which result can be altered image or report that is based on image analysis.

The continuum from image processing to computer vision can be broken up into low-, mid- and high-level processes

Low Level Process	Mid Level Process	High Level Process
Input: Image	Input: Image	Input: Attributes
Output: Image	Output: Attributes	Output: Understanding
Examples: Noise removal, image sharpening	Examples: Object recognition, segmentation	Examples: Scene understanding, autonomous navigation

Purpose of Image processing

The purpose of image processing is divided into 5 groups. They are:

- **Visualization** - Observe the objects that are not visible.
- **Image sharpening and restoration** - To create a better image.
- **Image retrieval** - Seek for the image of interest.
- **Measurement of pattern** – Measures various objects in an image.
- **Image Recognition** – Distinguish the objects in an image.

Methods used for Image Processing

Two methods are used for Image Processing are:

Analog image processing: In this type of processing, the images are manipulated by electrical means by varying the electrical signal. The common example include is the television image.

Digital image processing: It help in manipulation of the digital images by using computers. The three general phases that all types of data have to undergo while using digital technique are:

- Pre- processing
- Enhancement and display
- Information extraction

Example: Let 1 represents white and 0 represents black color.